Conserve mechanical energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Track kinetic and potential energy when nonconservative work is negligible.

SUCCESS CRITERIA - I can...

1. Identify the system and what your energy calculation treats as true.
2. Keep the total energy consistent between positions.
3. Explain how height and speed trade off.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mechanical energy” and “conserved” to explain your answer. Sentence starters are optional.

mechanical energy: total kinetic and potential energy in a model

conserved: unchanged in total under the stated conditions

system: the object or objects chosen for analysis

01 Warm-up: bring back an earlier idea

A **3 kg** object descends **5 m** near Earth.
1. Use g=**10**.
2. Find ΔU and the work done by gravity.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $K_i + U_i = K_f + U_f$; $K = \frac{1}{2}mv^2$; $U = mgh$.
Mechanical energy conserved when only conservative interactions transfer energy. Use g = **10 N/kg** near the surface.

Worked reasoning: A **1 kg** object drops **5 m** from rest with g=**10 m/s²** and no energy loss. It loses **50 J** of potential energy and gains **50 J** of kinetic energy, giving speed **10 m/s**.

Your turn: 1. **Draw energy bars** at the start and just before arrival.
2. **Label the system** and neglected losses.

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial **Transfers** **Final**

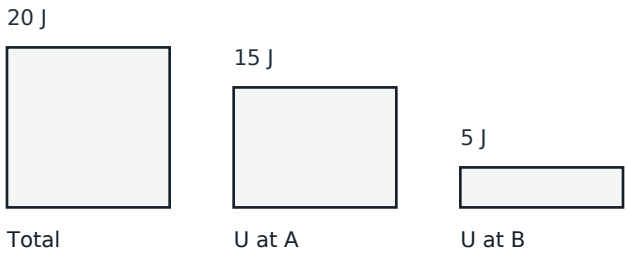
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A system has total mechanical energy **20 J**. At one point U=**15 J**; at another U=**5 J**. **Find K** at each point.

Sentence starter: I found … [unit]. This value means … in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **2 kg** ball starts at height **5 m** with speed **4 m/s**.
1. **Find its speed** at height **1 m**.
2. Use g = **10 N/kg** and ignore air resistance.
3. **Label the height reference** and **complete the energy account first**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A cart loses mechanical energy on a rough track. Does total energy conservation fail? **Name the destination** to include.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A and B have equal momentum magnitude, but A has **4** times B's mass. **Compare speeds** and kinetic energies algebraically.
Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: Both ... . They differ in ... because ... .

Before **After**

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A ball is thrown upward at **10 m/s** from h=**0**.
1. **Find maximum height** without drag, using g=**10**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **2 kg** cart starts at **4 m/s** and rises **3 m** without friction; g=**10**. Can it reach that height?
1. **Find its maximum rise**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3